Pol Mestres Ramon

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EDUCATION

University of California, San Diego — **La Jolla, USA**
PhD in Engineering Sciences. GPA 3.9/4 — *Expected graduation: 2025*

- Specialization in Dynamic Systems and Control
- Coursework: Linear Systems, Parametric Identification (Theory and Methods), Cooperative Control of Multiagent Systems, Nonlinear Systems, Optimal Estimation, Convex Optimization, Nonlinear Control, Linear Control Design, Optimal Control

University of California, San Diego — **La Jolla, USA**
Bachelor's Thesis. Grade: 10/10 MH — *September 2019 - March 2020*

- Title: Discretization of Optimization Flows through State-Triggered Control

Universitat Politècnica de Catalunya (UPC). — **Barcelona, Spain**
Degree in Mathematics. Average Grade: 8.34/10 — *September 2015 - May 2020*

Universitat Politècnica de Catalunya (UPC). — **Barcelona, Spain**
Degree in Engineering Physics. Average Grade: 8.25/10 — *September 2015 - May 2020*

Institut Jaume Vicens Vives — **Girona, Spain**
International Baccalaureate. Final mark: 39/45 — *September 2013 - June 2015*

Institut Jaume Vicens Vives — **Girona, Spain**
Batxillerat LOE. University Entering Mark: 9.45/10 — *September 2013 - June 2015*

PUBLICATIONS

Journal Publications

M. Vaquero, **P. Mestres**, J. Cortés, Resource-Aware Discretization of Accelerated Optimization Flows, IEEE Transactions on Automatic Control

RESEARCH AND WORK EXPERIENCE

University of California San Diego — **La Jolla, USA**
Graduate Student Researcher — *September 2020 -*

- Mathematical Modelling of Epidemics

Barcelona Supercomputing Center - Computational Biology Group — **Barcelona, Catalonia, Spain**
Research Intern — *June 2019 - Aug 2019*

- Data Science project in epigenetics. The aim of the project was to reconstruct a certain cell differentation tree by using epigenetic data: hi-c chromatine contacts, histone marks data, etc.

BaseTIS — **Barcelona, Catalonia, Spain**
Data Science Intern — *June 2018 - Aug 2018*

- Machine Learning project on attrition (predicting whether a given employee will leave a company)
- Deep Learning project on image classification for an olive oil company

Institut de Robòtica Industrial (IRI) — **Barcelona, Catalonia, Spain**

Summer Intern *June 2017 – Aug 2017*

- Worked on a project called "Detecting variable symmetries on continuous domains" with a researcher from the institute.

Honors

- Silver Medal at Catalan Physics Olympiad.
- Silver Medal at Spanish Physics Olympiad.
- Member of the winning team at Datathon CFIS 2018 on a Parkinson detection problem
- Finalist at HackUPC 2018 with a team that developed a tool to make simulations of the Barcelona Metro

Certifications

- TOEFL (score: 107/120)
- First Certificate in English
- HSK 1 title (Mandarin Chinese)

SKILLS

- Programming Languages: Python, C++, Matlab, R, SQL
- Other Software I've worked on: Latex, Ampl, LabView
- Native Catalan and Spanish, Fluent in English, Basic French and German.
- Chess ELO (FCE) around 2000